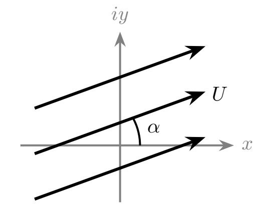
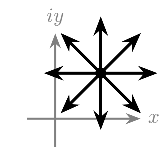
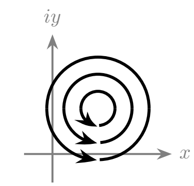
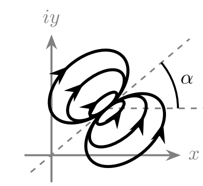
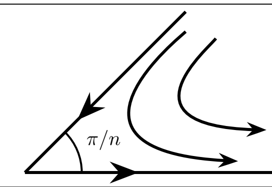
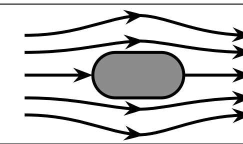
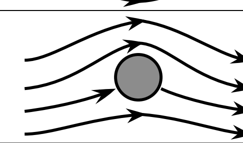
A Brief Handbook of Elementary Ideal Flows[1]

Elementary flow	**Complex potential**, $F = \phi + i\psi$	**Potential**, $\phi(x,y)$	**Streamfunction**, $\psi(x,y)$	**Sketch of streamlines**
1. Uniform flow at angle α to the horizontal	Az with $A = Ue^{-i\alpha}$ U is real but A is complex	$U(x\cos\alpha + y\sin\alpha)$ or $Ur\cos(\theta-\alpha)$	$U(y\cos\alpha - x\sin\alpha)$ or $Ur\sin(\theta-\alpha)$	
2. Source ($Q>0$; for sink replace Q by $-Q$) located at $z_0 = x_0 + iy_0$	$\dfrac{Q}{2\pi}\log(z-z_0)$	$\dfrac{Q}{4\pi}\ln\left[(x-x_0)^2+(y-y_0)^2\right]$ or, for $z_0=0$, $\dfrac{Q}{2\pi}\ln r$	$\dfrac{Q}{2\pi}\tan^{-1}\left(\dfrac{y-y_0}{x-x_0}\right)$ or, for $z_0=0$, $\dfrac{Q}{2\pi}\theta$	
3. Vortex (counterclockwise for $\Gamma>0$) located at $z_0 = x_0 + iy_0$	$\dfrac{-i\Gamma}{2\pi}\log(z-z_0)$	$\dfrac{\Gamma}{2\pi}\tan^{-1}\left(\dfrac{y-y_0}{x-x_0}\right)$ or, for $z_0=0$, $\dfrac{\Gamma}{2\pi}\theta$	$-\dfrac{\Gamma}{4\pi}\ln\left[(x-x_0)^2+(y-y_0)^2\right]$ or, for $z_0=0$, $-\dfrac{\Gamma}{2\pi}\ln r$	
4. Doublet located at $z_0 = x_0 + iy_0$ and rotated by angle α	$\dfrac{me^{i\alpha}}{z-z_0}$	$\dfrac{m\cos\left(\alpha-\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)\right)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$ or, for $z_0=0$, $\dfrac{m\cos(\alpha-\theta)}{r}$	$\dfrac{m\sin\left(\alpha-\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)\right)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$ or, for $z_0=0$, $\dfrac{m\sin(\alpha-\theta)}{r}$	
5. Flow in a corner[2] of angle π/n	Az^n	... or $Ur^n\cos(n\theta)$	... or $Ur^n\sin(n\theta)$	
6. Flow past a Rankine body (a is a real number/horizontal shift) 1$\{\alpha=0\}$ + 2$\{z_0=-a\}$ − 2$\{z_0=a\}$	$Uz + \dfrac{Q}{2\pi}\log\left(\dfrac{z+a}{z-a}\right)$	$Ux + \dfrac{Q}{4\pi}\ln\left[\dfrac{(x+a)^2+y^2}{(x-a)^2+y^2}\right]$	$Uy + \dfrac{Q}{2\pi}\left[\tan^{-1}\left(\dfrac{y}{x+a}\right) - \tan^{-1}\left(\dfrac{y}{x-a}\right)\right]$	
7. Flow past a cylinder with circulation 1$\{\alpha=0\}$ + 4$\{\alpha=0, m=UR^2, z_0=0\}$ − 3$\{z_0=0\}$	$U\left(z+\dfrac{R^2}{z}\right) + \dfrac{i\Gamma}{2\pi}\log z$	... $U\left(1+\dfrac{R^2}{r^2}\right)r\cos\theta - \dfrac{\Gamma}{2\pi}\theta$	... $U\left(1-\dfrac{R^2}{r^2}\right)r\sin\theta + \dfrac{\Gamma}{2\pi}\ln r$	

[1] Inspired by Tables 12.1 and 12.2 of Granger's *Fluid Mechanics* (Dover Publications, 2012). Flow 1–5 are the "building blocks" and the rest are superposition examples.
[2] The case $n=2$ corresponds to stagnation-point flow onto a horizontal plate.